

SHAMAL MUSTHAFA

Data Scientist | GenAI / LLM Engineer | AI Engineer | Data Engineering | Business Intelligence

Madurai, Tamil Nadu, India | +91 7012464811 | shamalmusthafa59@gmail.com
linkedin.com/in/shamalmusthafa | github.com/Shamal119 | shamal-portfolio.vercel.app

PROFESSIONAL SUMMARY

Data Scientist and AI Solutions Builder with 2+ years of experience building Generative AI applications, RAG knowledge systems, LLM chatbots, backend AI services, cloud ETL pipelines, BI dashboards, machine learning models, and workflow automation. Strong hands-on experience with Python, FastAPI, LangChain, FAISS, Azure OpenAI, OpenAI API, Gemini API, AWS Glue, Amazon S3, Redshift Serverless, SQL, Power BI, Tableau, and Alteryx. Known for converting business requirements into production-oriented AI tools, analytics pipelines, dashboards, and automation workflows with measurable operational impact.

CORE TECHNICAL SKILLS

Programming & Data Science: Python, SQL, Pandas, NumPy, scikit-learn, XGBoost, feature engineering, EDA, classification, regression, anomaly detection, model evaluation

Generative AI / LLM Engineering: RAG, LangChain, FAISS, vector search, embeddings, chunking, semantic search, prompt engineering, prompt templates, conversational AI, AI agents, tool calling, MCP

AI Platforms & NLP: Azure OpenAI, OpenAI API, Gemini API, Vertex AI, Dialogflow CX, OpenAI Whisper, NLP, transcript analytics, text summarization, information retrieval

Backend / Full Stack: FastAPI, REST APIs, NestJS, React, Next.js, TypeScript, JavaScript, Streamlit, API integration, authentication, OTP workflows, SQLAlchemy

Cloud Data Engineering: AWS Glue, Amazon S3, Redshift Serverless, Amazon RDS, MySQL, PostgreSQL, SQL Server, ETL/ELT, incremental loads, watermark logic, data migration, validation, reconciliation

BI / Analytics / Automation: Power BI, Tableau, DAX, Alteryx Designer, dashboarding, KPI reporting, data modeling, workflow automation, Git, Docker basics, Agile delivery

PROFESSIONAL EXPERIENCE

Data Scientist / AI Solutions Builder | Truwave Software LLC, Madurai | Aug 2023 - Present

- Built enterprise GenAI and LLM solutions using Azure OpenAI, OpenAI/Gemini APIs, LangChain, FAISS, and Python for RAG, knowledge search, chatbot workflows, and AI-assisted information access.
- Designed RAG knowledge base systems with document ingestion, chunking, embeddings, vector retrieval, prompt templates, and conversational response generation for context-aware Q&A.
- Developed backend AI services in FastAPI and NestJS for chatbots, intelligent search, file processing, ticket summarization, OTP flows, and API-based automation integrated with React/Next.js applications.
- Created Gemini-powered ticket summarization and support-review workflows from structured ticket data, reducing manual review effort by about 40% and improving triage speed.
- Built AWS analytics pipeline from MySQL RDS to AWS Glue, Amazon S3, Redshift Serverless, and Power BI with transformation layers and watermark-based incremental loading, reducing downstream reporting data movement by about 99%.
- Designed SQL/Python data migration workflows using raw, staging, validation, and target layers with legacy ID preservation, ID mapping, parent-child dependency checks, reconciliation, and rollback planning.
- Delivered Power BI and Tableau dashboards for billing, meter usage, call-center analytics, and operational KPI reporting with clean data models, DAX measures, filters, drilldowns, and stakeholder-ready views.
- Automated recurring data preparation and reporting workflows with Python, SQL, and Alteryx Designer, reducing processing effort by about 30% while improving repeatability of analytics delivery.
- Built water-meter usage analytics across hourly, daily, monthly, and yearly granularity with quality checks, rolling features, anomaly detection, and leak/no-usage detection logic to support low-false-positive alerting.

Data Analyst Intern | Truwave Software LLC, Madurai | Jan 2023 - Apr 2023

- Cleaned, transformed, and analyzed call-center and operational datasets using Python, SQL, Alteryx, and BI workflows for reporting and performance analysis.
- Built Power BI dashboards to track call performance, customer satisfaction signals, service trends, and operational KPIs.
- Explored transcript analytics using OpenAI Whisper and NLP techniques to convert unstructured call data into measurable insights and reporting-ready features.

SELECTED PROFESSIONAL PROJECTS

- **Enterprise RAG Knowledge Base:** Built retrieval-grounded AI assistant over business knowledge sources with document ingestion, embeddings, vector search, prompt workflows, and conversational Q&A.; Tech: Python, LangChain, FAISS, Azure OpenAI/OpenAI, FastAPI.
- **AI Ticket Summarization System:** Automated support-ticket summarization and search workflows that turn long ticket history into concise, actionable summaries. Tech: Gemini API, NestJS, REST APIs, SQL.
- **AWS ETL & BI Pipeline:** Created cloud data pipeline from MySQL RDS to AWS Glue/S3 to Redshift Serverless and Power BI with transformation logic, incremental loads, and reporting-ready output tables. Tech: AWS Glue, S3, Redshift, RDS, SQL, Power BI.
- **Water Leak / Usage Anomaly Detection:** Built meter analytics with rolling features, no-usage checks, anomaly scoring, and data-quality rules to support leak detection. Tech: Python, Alteryx, Power BI, SQL, anomaly detection.
- **Legacy Data Migration Framework:** Designed raw-to-staging-to-validation-to-target migration process with preserved legacy IDs, mapping tables, referential-integrity validation, reconciliation, and sign-off checks. Tech: Python, SQL, ETL, data validation.

INDEPENDENT PROJECTS & AI AUTOMATION

- **Customer Churn Prediction App:** Developed an end-to-end churn prediction solution with feature engineering, XGBoost model evaluation, FastAPI serving, and a lightweight React frontend; achieved 84.6% ROC AUC. Tech: Python, XGBoost, FastAPI, React, Vercel.
- **Pipe Failure Prediction / Predictive Maintenance:** Built predictive analytics prototypes using asset, usage, and operational features to support risk ranking, anomaly detection, and maintenance prioritization. Tech: Python, scikit-learn, SQL, Power BI.
- **MCP Demo Server:** Built a FastMCP demo server with simulated Jira, GitHub, and CloudWatch-style data to demonstrate tool-based AI querying and agentic workflow patterns. Tech: FastMCP, Python, tool calling.
- **Next.js / FastAPI Auth System:** Built login and registration workflows with OTP-style authentication, backend APIs, database models, and frontend pages for app onboarding. Tech: Next.js, FastAPI, SQLAlchemy, PostgreSQL.
- **Computer Vision Research:** Presented CIDA-23 research on side-face image generation for vulnerability analysis of face-recognition algorithms and dataset creation for robustness evaluation. Tech: computer vision, face recognition, dataset generation.

EDUCATION

M.Sc. Data Science | CHRIST University

Bachelor of Computer Applications (BCA) | Kannur University

CERTIFICATIONS & RESEARCH

- Alteryx Designer Core Certified
- Alteryx Foundational Micro-Credential
- Presented research paper: Side Face Image Generation for Vulnerability Analysis of Face Recognition Algorithms at CIDA-23 national conference on Computational Intelligence and Data Analytics.
- Microsoft AI learning: Fundamental AI Concepts / Azure AI learning exposure
- Modern NLP using Deep Learning, Udemy

ADDITIONAL TECHNICAL KEYWORDS

Data Scientist, Generative AI Engineer, LLM Engineer, AI Engineer, Machine Learning Engineer, Python Developer, FastAPI, LangChain, RAG, Azure OpenAI, OpenAI API, Gemini API, FAISS, vector database, embeddings, semantic search, prompt engineering, AI agents, MCP, NLP, XGBoost, scikit-learn, anomaly detection, predictive modeling, AWS Glue, Amazon S3, Redshift Serverless, SQL, ETL, data migration, Power BI, Tableau, DAX, Alteryx, automation, REST APIs, React, Next.js, NestJS.